

MID SEMESTER EXAMINATION, AUTUMN-2024

School of Applied Sciences, KJIT-DU

Course: Basic Instrumentation

Code: EE 10003

Full Marks: 20

Time: 90 minutes

Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

Q.1		Marks	CO
a)	Name the three torques required for proper operation of the indicating instruments.	1	1
b)	Why the scale of the spring control type instruments is uniform?	1	1
c)	What is the importance of the soft iron core in the PMMC instruments?	1	1
d)	How the current coils, and pressure coils are connected for measurement of power using two-wattmeter method?	1	2
e)	What is the electrical displacement in degrees between different phases of a three-phase generator?	1	2

Q.2		Marks	CO
a)	The deflecting torque of an instrument is directly proportional to the current passing through it, and the instrument has full scale deflection of 70° for a current of 20 A. What deflection will occur for a current of 10 A when the instrument is (i) spring-controlled (ii) gravity controlled?	3	1
b)	Why damping torques is essential in the indicating type instruments? What is the role of balance weight in the gravity control method for controlling the torque of an indicating instrument?	2	1

Q.3		Marks	CO
a)	Discuss the construction and working principle of a permanent magnet moving coil instrument with a neat schematic diagram.	3	1
b)	The coil of a permanent-magnet moving coil instrument has 20 turns on a rectangular former of $3.5 \text{ cm} \times 1.5 \text{ cm}$ and swings in a uniform field of 0.18 Wb/m^2 . If a steady current of 50 mA is flowing through the coil, calculate the deflecting torque.	2	1

Q.4		Marks	CO
a)	Derive the torque equation of a moving iron instrument.	3	1
b)	Find the deflection of moving iron type ammeter using following data: Control spring constant = $16 \times 10^{-6} \text{ Nm/rad}$; Current = 20A; $L = 30 + 5\theta - 2\theta^2 \mu \text{H}$	2	1

Q.5		Marks	CO
a)	Two wattmeter method is used to measure the power absorbed by a 3-phase induction motor. The wattmeter readings are 12.5 kW and -4.8 kW. Find (i) the power absorbed by the machine, (ii) Load power factor, and (iii) reactive power taken by the load.	3	2
b)	Draw the schematic diagram for power measurement in a 3-phase circuit by two wattmeter method. What is the difference between phase voltage, and line voltage in a 3-phase star connected circuit. Can we measure power in 3-phase circuit by two wattmeter method if the loads are unbalanced?	2	2

Best of Luck
